

• THE RAT PATROL

EST. DIVHACKS · MMXXVI

NightOwl

Nothing hides from the Owl.

Finding the rats New York isn't looking for.

PITCH NO.

01 / NIGHTOWL

RUNTIME

~3 MINUTES

VENUE

DIVHACKS · COLUMBIA

CREW

ETHAN · BRUNO · UTSAV · SANJAVAN



01 · THE PROBLEM

New York's rat map is a map of *who calls*.

High-income, English-speaking blocks call the most.

76%

more complaints from the rich than the poor, for the same number of rats.

RICHEST VS POOREST FIFTH OF NYC BLOCKS · 2015-2026

51%

more rat infestations on poor blocks than on rich ones.

SWEPT LOTS WITH ACTIVE RATS · 12.2% VS 8.1%

— EXHIBIT A · BY INCOME

Quieter phones. *More rats.*

	West Village	East Harlem (South)
Median household income	\$188k	\$39k
Poverty rate	6%	31%
Homes with limited English	1%	16%
Rat complaints to 311 PER 1,000 RESIDENTS, A YEAR	5.7	3.2
Swept lots with rats ROUTINE BLOCK SWEEPS	6.4%	15.9%

2.5× the rats, 44% fewer calls

EXHIBIT B · BY LANGUAGE, SAME INCOME

Quieter phones. *More rats.*

	Crown Heights (South)	Jackson Heights
Median household income	\$84k	\$81k
Poverty rate	17%	13%
Homes with limited English	5%	25%
Rat complaints to 311 PER 1,000 RESIDENTS, A YEAR	5.0	2.2
Swept lots with rats ROUTINE BLOCK SWEEPS	4.7%	7.9%

Same income. *1.7× the rats, 56% fewer calls*

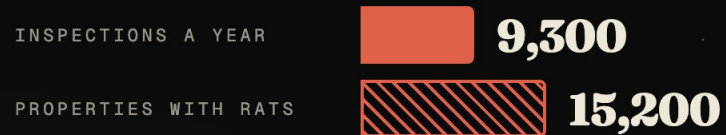
THE SCALE

~150,000 rat inspections a year.
They go where the calls are.

• *Richer blocks that call more*

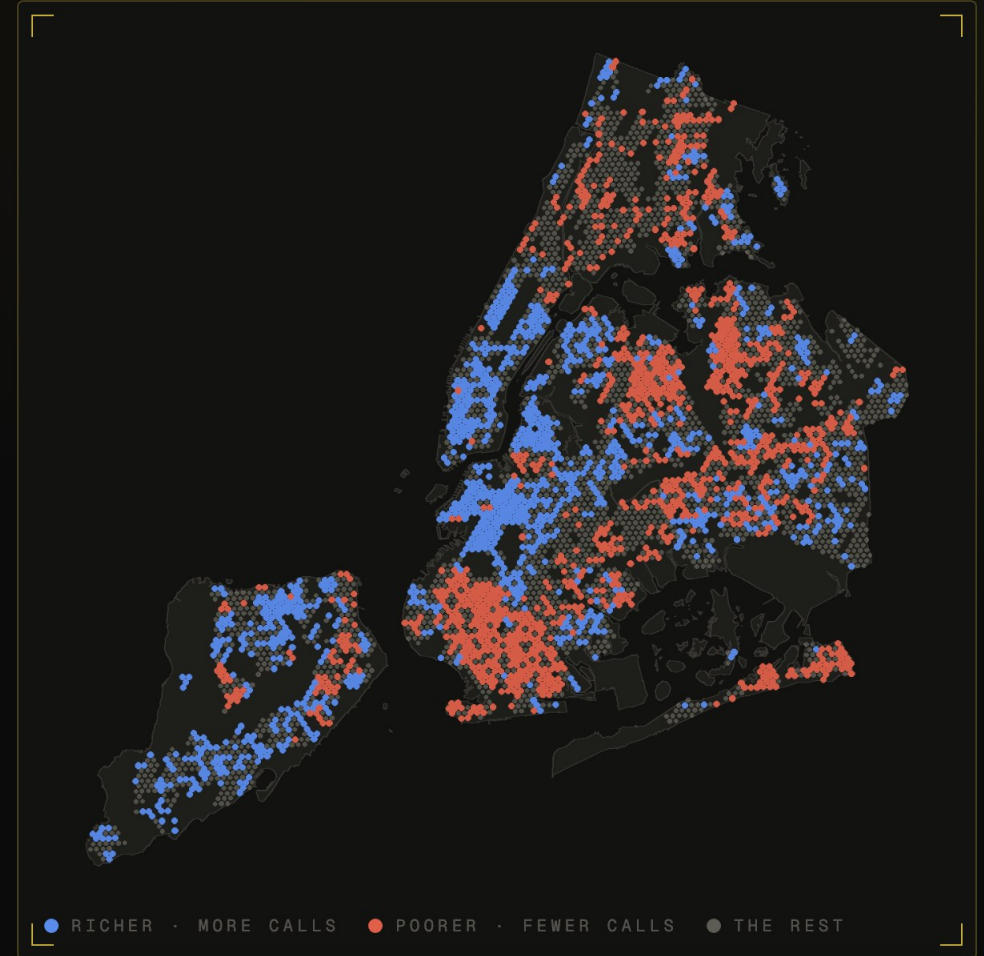


• *Poorer blocks that call less*



Blue gets 5× the inspections for 1.4× the rats.

...and it leaves 382 blocks behind.



— THE SILENT BLOCKS

382

blocks where the rats are likely and
the *phones are quiet*.

SILENT MEANS ALL FOUR

- i.* Rat risk in the **top 40%** of NYC blocks (Model B)
- ii.* Its 311 calls per resident rank at least **25 points** (of 100) below its rat risk
- iii.* All **5 of 5** model copies agree
- iv.* At least **100 homes**, so it's not a park or rail yard

WHAT THEY LOOK LIKE

1.2M

people live on them

2×

NYC's average rat risk, with under half its calls per resident

\$62k

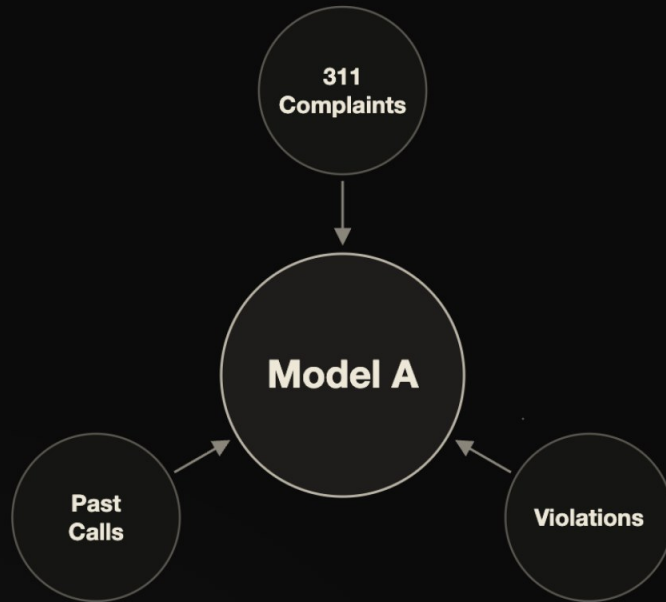
median income (NYC: \$90k)

17%

homes with limited English (NYC: 10%)

02 · THE SOLUTION

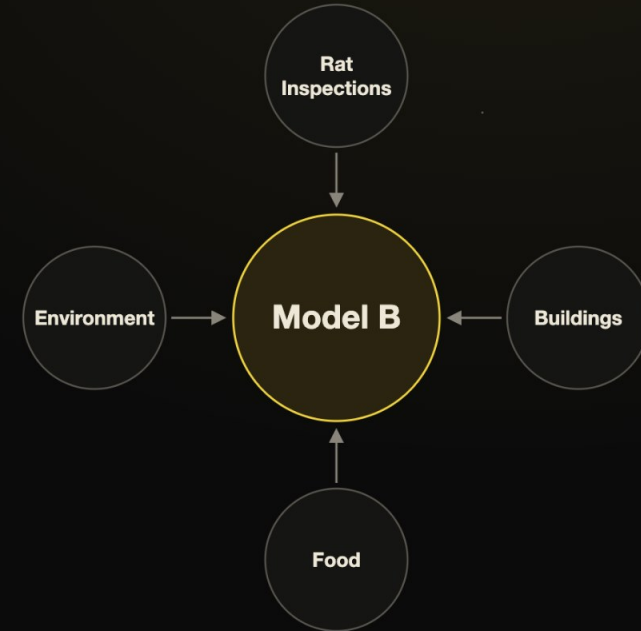
Two models, *one gap*.



MODEL A · TRAINED ON 311 CALLS

What the city hears

VS



MODEL B · TRAINED ON EVERYTHING OBJECTIVE

What's actually there

B high, A low: a *silent block*.

— THE PROOF

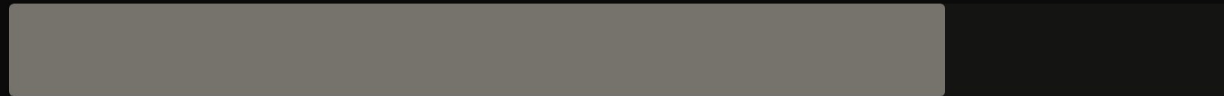
Ten years of history. Who finds *more rats*?

Model B's picks



19.5%

Following 311 calls



15.2%



THE OWL

Meet the *Owl*.

A sensor that spots rats in any light, every night.

i.

COSTS

About **\$30**, one inspector-hour.

ii.

LASTS

Up to **6 weeks** on watch, every night.

iii.

PRIVATE BY DESIGN

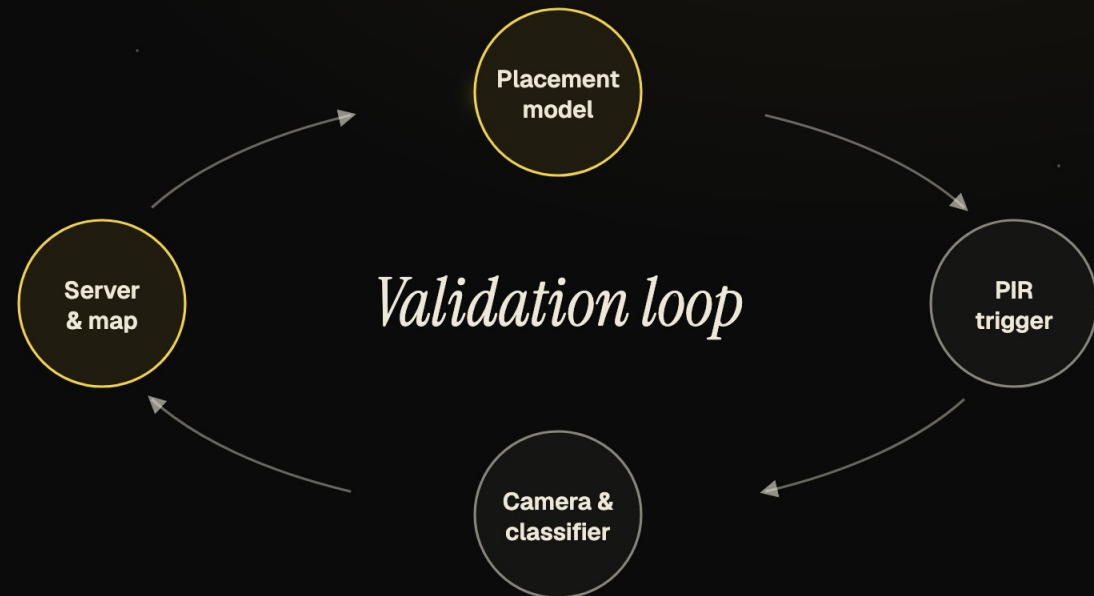
Classification runs on the device. No video leaves it: only a rat event (time, block, confidence) and a crop of the confirmed rat.



— THE OWL

Meet the *Owl*.

A sensor that spots rats in any light, every night.



Rats are a *warning light*.

Rat infestations

HEALTH

• **24**

Leptospirosis cases in NYC in 2023, a record. About 3 a year before 2021.

NYC HEALTH ADVISORY, 2024

• **1 in 3**

Asthma inner-city homes had rat allergen, linked to worse asthma in kids.

INNER-CITY ASTHMA STUDY, 2003

• **72%**

Depression more likely where rats are a big problem.

JOHNS HOPKINS · BALTIMORE, 2016

HOME & STREET

• **2.8×**

Mold the mold violations per home.

NYC HPD · NYC HEALTH SWEEPS · 2023-25

• **2.3×**

No heat the heat and hot-water complaints per home.

NYC 311 · 2023-25

• **1.8×**

Illegal dumping the dumping complaints per property.

NYC 311 · 2023-25

Then: the same method for *mold, heat outages and dumping*, anything the city only hears about through calls.

Try it *yourself.*

— THE CLOSE —

Nothing hides from the *Owl*.

*NightOwl turns silence into data,
and data into the next place to look.*

• THANK YOU

DIVHACKS 2026

NYC, BLOCK BY BLOCK

Four humans, *one* Owl.



Ethan

I · MAP & API



Bruno

II · HARDWARE & ASSISTANT



Utsav

III · RAT DETECTOR



Sanjavan

IV · MODELS & DATA

The tools that let us *see in the dark*.

MODELS · HERO

LightGBM × 2

Two gradient-boosted models, ten years of NYC data, tested on held-out districts.

MAP GRID

H3 hexagons

~350 m cells, 2–3 blocks

CITY DATA

NYC Open Data

3.1M inspections · 512K complaints

DETECTOR

Rat detector V5

object detector · ~15 fps on the Pi

NODE

Raspberry Pi 5

NoIR camera · PIR sensor

API

FastAPI

events · scores · Owl sites

MAP

React · three.js

3D city, block by block

ASSISTANT

Grok · xAI

20 tools, plain English

MESSAGING

Photon

the assistant on iMessage

The backtest, *month by month.*



- 119 months, 2016–2026, each trained only on the past.
- Scored on **routine sweeps**, not complaint visits.
- Model B **19.5%** vs 311 **15.2%**: won **108 of 119** months.
- Quiet blocks: **15.7%** vs 8.8% for random picks.

LIMIT: COVERS BLOCKS THE CITY SWEEP. IT IS NOT A CITYWIDE RAT COUNT.

Raw city data, *no model*.

- Per rat found, the richest fifth of blocks calls **1.8×** as much as the poorest, and the most English-fluent fifth **2.2×** the least.
- Same high rat risk: quiet blocks got **29** inspections per 100 properties in two years, loud ones **145**.
- Routine sweeps do reach the poorest fifth most. The blind spot is **poor and quiet**.
- Research agrees: low-income, minority areas report less despite greater need (Kansas City, 2021).

Bytes, *not video*.

-
- Frames stay on the Owl. By design, nothing is streamed or stored as video.
 - An event is a time, a block, the class, and a confidence, plus a cropped photo only for a confirmed rat.
 - The motion sensor wakes the camera. Otherwise it sleeps.
 - Only rat events leave the Owl, never people.
 - Unlike Flock or ShotSpotter-style systems, it records no people and no plates. It counts rats.

About *\$30* an Owl.

- **Demo Owl:** Raspberry Pi 5, IMX219 NoIR night camera, HC-SR501 motion sensor, 850 nm IR light, 3D-printed case on a gooseneck mount.
- Motion wakes the camera. The detector runs on the device, about 15 frames a second.
- **Production design:** the same detector on a ~\$14 ESP32-S3 camera board, about **\$30 per Owl**.
- **Up to 6 weeks** on watch applies to the low-power production design, not the Pi 5 demo Owl.
- It mounts on street-tree guards, and the planner picks the exact tree.